



FROM KITS TO CORTEX

Calibrating a Neurocognitive Protocol for Hands-On Physics
Education in Under-Resourced Classrooms

Gabriel Moreno Ribeiro

Projeto Candela

Former PIBIC Jr. Fellow, Universidade Federal da Bahia
Colégio Militar de Salvador
Salvador, Bahia, Brazil

FUNDED BY



Programa Institucional de Bolsas de Iniciação Científica Júnior
(PIBIC Jr./CNPq)

PRESENTED AT



Instituto Internacional de Física, selected among 14 projects
from over 700 submissions

Institutional marks appear with written authorization and denote affiliation,
funding, or presentation venue. They do not imply institutional endorsement
of the findings reported here.

Preprint, August 6, 2026

Data Availability

All code, data, and analysis scripts are available in the project repository. The repository contains three layers of data, clearly separated: (1) real implementation data from 28 schools (L1), (2) effect sizes extracted from published literature with verified DOIs (L2), and (3) simulated neural data calibrated on the pooled L2 estimates (L3). No raw neural data exists; this study is a protocol with design analysis.

Funding

This work was supported by the Programa Institucional de Bolsas de Iniciação Científica Júnior (PIBIC Jr./CNPq), Universidade Federal da Bahia.

Ethics Statement

No data collection involving human participants has been conducted for this study. The manuscript presents a protocol and simulation. Ethics approval from the Comitê de Ética em Pesquisa (CEP) and CONEP will be sought prior to any deployment involving minors. Parental informed consent and child assent procedures are described in the protocol section.

Conflict of Interest

The author founded and distributes the Projeto Candela intervention evaluated in this study. This relationship is disclosed in full. The analytic pipeline is fully reproducible and all priors are derived from independently published literature, allowing independent verification of all claims.

Author Contributions (CRediT)

Gabriel Moreno Ribeiro: Conceptualization, Methodology, Investigation (implementation), Writing (Original Draft), Visualization, Funding Acquisition, Project Administration.

Acknowledgments

The author thanks Prof.^a Dr.^a Maria das Graças (UFBA) for research mentorship; Prof. Jorge Lúcio, Cel. Iran, and Cel. Magalhães (Colégio Militar de Salvador) for institutional support and didactic guidance; the crowdfunding backers who supported kit production; and the 28 partner schools in Bahia and Ceará that participated in the implementation phase.

Abstract

Low-cost science kits are deployed at scale across the Global South yet evaluated almost exclusively by test scores, which cannot distinguish which instructional modality drives learning gains. We present a neurocognitive protocol designed to fill this gap. First, we deployed five hands-on physics experiments across 28 public schools in Bahia and Ceará, Brazil (3,392 students, R\$83 per kit). Second, we conducted a systematic synthesis of neuroeducation effect sizes across 13 primary EEG and fNIRS studies ($k = 13$, pooled Hedges' $g = 0.640$, deflated to $g = 0.414$ after PET bias correction). Two additional meta-analyses (Freeman et al., 2014; Chikhi et al., 2022) were retained as contextual references but excluded from primary pooling to avoid double-counting. Third, we calibrated a Bayesian design analysis on these pooled posteriors to determine the sample size, measures, and analytic plan required to distinguish four instructional conditions (lecture, video demonstration, live demonstration, and student-performed experimentation) using portable neurophysiological recording. This paper reports three data layers: real implementation data (L1), effect sizes from published literature (L2), and simulated neural outcomes calibrated on L2 (L3). No neural data has been collected. All simulated numbers are labeled as such. The simulation yields a minimum of 100 students for 80% frequentist power to detect the C3→C4 contrast (simulated $g = 0.249$; Type S error < 0.001, Type M ratio = 1.18 at $n = 80$). The specification-curve analysis across 243 analytic variants shows a stable point estimate (median $g = 0.248$, range [0.240, 0.259]) robust to researcher degrees of freedom, though significance at $n = 30$ is 0% as expected from the power analysis. We provide a pre-specified falsification criterion: if the C4 (student-performed) condition fails to exceed C3 (live demonstration) by at least $g = 0.15$ with non-overlapping 89% credible intervals, the Candela enactive hypothesis is refuted.

Keywords: neuroeducation, fNIRS, EEG, embodied cognition, hands-on learning, design analysis, Bayesian meta-analysis, low-cost science education, Brazil

1 Introduction

Hands-on science education is not new. What remains unresolved is the question of *why* it works at the neural level, and whether the specific mech-

anism (sensorimotor engagement, sustained attention, or social co-regulation) can be isolated under field conditions. This paper tries to address that question for a concrete case: five low-cost physics experiments deployed across 28 Brazilian public schools, serving 3,392 students at a total cost of R\$83 per kit.

The contribution is threefold, in order of novelty:

1. A harmonized synthesis of neuroeducation effect sizes across instructional modalities ($k = 13$ primary studies with neural measures, multiple neuroimaging techniques), converted to a common Hedges' g scale. This conversion appendix, which places fNIRS oxygenation changes and EEG band-power ratios on the same metric, allows cross-paradigm comparison that test-score studies cannot provide. Two widely-cited meta-analyses (Freeman et al., 2014; Chikhi et al., 2022) are retained as contextual references but excluded from primary pooling to avoid double-counting.
2. A Bayesian design analysis that specifies what sample size a real school-based deployment needs, including Type S and Type M error rates in addition to statistical power. The simulation is calibrated on deflated posteriors from the synthesis, after explicit bias correction for the known small-study problem in neuroeducation.
3. A cost-calibrated feasibility argument: what it costs, in Brazilian reais, to detect each unit of neural effect in a classroom with no laboratory bench, intermittent power, and 50-minute class periods.

1.1 The Problem

Test scores confirm that something changed after a hands-on intervention. They cannot distinguish engagement from memorization; they cannot identify which component of the instructional experience drove the change; and they cannot inform decisions about resource allocation at the level a Secretaria de Educação requires. Neurophysiological measures (prefrontal oxygenation, frontal theta power, sensorimotor mu desynchronization, inter-brain synchrony) can, in principle, make these distinctions. The studies we found on these measures run predominantly on university students, in laboratory settings, in high-income countries. None of them had been run under the constraints of

a Brazilian public school with no dark room, no research technician, and one 50-minute period.

1.2 The Gap

Three literatures seem to converge on this question, though we did not find work that connects them directly:

1. **Inquiry-based science outcomes.** A large meta-analytic base shows that active learning improves exam scores by approximately $g = 0.47$ relative to traditional lecture [1]. The evidence establishes *that* it works; it does not address *how*.
2. **Embodied cognition.** Theoretical and empirical work establishes that sensorimotor engagement during learning activates distinct neural circuits (motor cortex and dorsolateral prefrontal regions) beyond those recruited by passive observation [2]. This work is conducted with adults in laboratories.
3. **Ecological neuroeducation.** A small but growing body of work uses portable EEG and fNIRS in real classrooms [3, 4]. Sample sizes are small, paradigms vary, and no study has yet mapped the four-condition gradient from passive lecture through enactive manipulation that a kit-based intervention naturally provides.

This paper bridges these literatures. It does not collect neural data (that has not happened and this limitation is stated upfront), but synthesizes the existing evidence, calibrates a simulation on it, and specifies the exact protocol, sample size, and falsification criterion that a future deployment requires.

1.3 Structure of This Paper

Section 2 reviews the three convergent literatures. Section 3 describes the Candela intervention and its five experiments. Section 4 presents the systematic synthesis of neuroeducation effect sizes. Section 5 specifies the four-condition protocol and its directional hypothesis, including measurement instruments. Section 6 describes the analysis plan. Section 7 reports the Bayesian design analysis, Type S/M errors, and specification-curve results. Section 8 presents the simulated neural outcomes. Section 9 bridges cost and effect. Section 10 discusses implications. Section 11 leads with the central limitation: no neural data has been collected.

A note on authorship

This paper was written by a secondary school student during a PIBIC Jr./CNPq fellowship at the Universidade Federal da Bahia, under faculty supervision. The project grew from five physics experiments built and tested in 28 public schools in Bahia and Ceará. No neuroimaging equipment was available at any point, which is precisely why this paper is a protocol and a simulation rather than a data report. The synthesis, the design analysis, and the cost argument were developed to answer a practical question: if a future researcher wanted to run this study properly, what would they need to know before starting?

2 Related Work

Three research threads informed this protocol. Reading them made clear what the present design needed to add.

2.1 Inquiry-Based Science Outcomes

The meta-analysis by Freeman et al. (2014) pooled 225 studies comparing active learning to traditional lecturing in undergraduate STEM courses. The pooled effect on exam scores was $g = 0.47$ (95% CI [0.38, 0.56]), and students in lecture-only sections were 1.5 times more likely to fail, a finding that is robust and well-powered.

What it does not provide: a mechanism. “Active learning” encompasses peer discussion, clicker questions, group problem-solving, and hands-on manipulation. These differ in the sensorimotor, attentional, and social demands they place on the student. A test score cannot distinguish which demand drove the gain.

Deslauriers et al. (2019) added a nuance: students in active classrooms learned more ($d = 0.33$) but *perceived* learning less, because cognitive effort feels like failure to novices. This perception gap matters for adoption in contexts where teacher evaluation relies on student satisfaction.

What we did not find: We did not find a meta-analysis that decomposes “active learning” into its constituent modalities with a shared neural metric. Test-score studies do not provide this kind of mechanistic breakdown.

2.2 Embodied Cognition and Sensorimotor Learning

The embodied cognition framework predicts that physical manipulation during learning recruits sensorimotor cortex in a way that observation alone does not. Cannon et al. (2014) demonstrated this with EEG: adults who *performed* a novel tool action showed greater mu rhythm desynchronization (8–13 Hz) during subsequent observation than adults who had only watched the action. The effect was lateralized to the right hemisphere and was large in the performer group relative to novices.

Hobson and Bishop (2016) qualified this finding with the largest mu suppression study to date ($n = 61$). While execution produced strong and reliable mu suppression, the observation effect was “weak and unreliable” and 16–21% of participants showed no mu suppression at all. This study is a corrective to uncritical claims about the mirror neuron system.

Van der Weel and Van der Meer (2024) provided a high-density EEG (256 channels) comparison of handwriting versus typewriting in 36 university students. Handwriting produced widespread theta/alpha coherence in parietal and central regions. This is the closest analogue to the kit-versus-video contrast in the Candela design: physical manipulation versus a less motorically demanding modality, with the same content.

What we did not find: Nearly all embodied cognition neuroimaging we encountered was conducted with adults in laboratories. We did not find studies that tested this with children in noisy classrooms, so the translation to that setting remains an open question. The present protocol is designed to examine it.

2.3 Ecological Neuroeducation

Dikker et al. (2017) recorded portable EEG from 12 high school students simultaneously over 11 class sessions. Brain-to-brain synchrony across students predicted engagement ratings and tracked social dynamics. This study established that classroom EEG hyperscanning is technically feasible.

Pan et al. (2020) used fNIRS hyperscanning with instructor-learner dyads and found that brain-to-brain coupling was present during scaffolding (active/constructive instruction) but absent during explanation (passive/didactic instruction). Coupling predicted learning outcomes. This is the closest published precedent to the four-condition contrast we propose: the neural signature of in-

structional modality, measured in real time.

More recent work by Mendoza-Armenta et al. (2024) and Li et al. (2024) confirms that portable EEG and fNIRS can capture teacher-student neural coupling in educational contexts with sample sizes of 20–40. Ko et al. (2017) and Aghajani et al. (2017) further demonstrate that EEG+fNIRS can resolve parametric engagement differences in real learning environments. Dikker et al. (2021), Apicella et al. (2022), Zhu et al. (2021), and Causse et al. (2017) contribute additional evidence that portable neuroimaging captures cognitive engagement across diverse conditions.

What we did not find: Sample sizes ranged from $n = 12$ to $n = 100$ (Dikker et al., 2021, in a museum setting), with most classroom studies below $n = 40$. We could not find a study that compared four graduated conditions (passive reception through full enactive manipulation) within the same classroom using the same content. Effect sizes were rarely reported in a form that permitted cross-study comparison, which is part of what motivated the synthesis in Section 4.

3 The Candela Intervention

Projeto Candela is a set of five physics experiments designed for Brazilian public schools that lack laboratory infrastructure. Each experiment uses materials costing between R\$8 and R\$30 (total kit cost: R\$83, approximately USD 16 at 2023 exchange rates). The project was developed at the Colégio Militar de Salvador under a PIBIC Jr./CNPq fellowship at the Universidade Federal da Bahia and deployed across 28 schools in the states of Bahia and Ceará, reaching 3,392 students.

3.1 The Five Experiments

Table 1 summarizes the five experiments. Each targets a specific physics misconception, maps to a competency in the Base Nacional Comum Curricular (BNCC), and involves a concrete enactive component: the hands-on manipulation that distinguishes condition C4 from C1–C3 in the proposed protocol.

3.2 Design Principles

Three principles governed kit design:

1. **Cost constraint.** No experiment may require materials exceeding R\$30 total. This excludes commercial lab equipment but not common

Table 1: Summary of the five Projeto Candela experiments (28 schools, Bahia and Ceará).

ID	Experiment	Physics domain / BNCC	Target misconception	Enactive component	Cost
1	Avg. velocity	Kinematics / EF09CI01	Confusing speed with velocity; ignoring direction	Student times car, marks positions, computes $v = \Delta d / \Delta t$	R\$15
2	Friction coeff.	Dynamics EM13CNT301	/ Friction depends on contact area; confusing μ_s and μ_c	Varies ramp angle, measures $F_{\sin \theta}$, isolates $\mu = F_{\text{at}} / N$	R\$18
3	Hooke’s law	Elasticity EM13CNT301	/ Proportionality only for ideal springs; conflating k with stiffness	Hangs masses vertically and on inclined plane, measures Δx , computes k	R\$12
4	Pendulum (g)	Oscillations EM13CNT301	/ Period depends on mass or amplitude (small θ)	Varies L and θ , times oscillations, solves $g = 4\pi^2 L / T^2$	R\$8
5	Buoyancy	Fluid statics / EF09CI01	Buoyancy depends on weight not displaced volume	Immerses object, marks level, computes $\varepsilon = \rho V g$, $P_a = mg - \varepsilon$	R\$30

Total kit cost: R\$83 ($\approx$ USD 16 at 2023 exchange rates). All experiments conducted at Colégio Militar de Salvador under PIBIC Jr./UFBA, 2023. Each session: 50 min.

retail items (toy cars, springs, string, water containers).

- 2. Single-period constraint.** Every experiment fits within one 50-minute school period, including setup, execution, data collection, and minimal cleanup.
- 3. Misconception targeting.** Each experiment directly confronts a documented student misconception (e.g., “pendulum period depends on mass”, “friction depends on contact area”). The enactive component is designed so that the physical experience contradicts the misconception.

3.3 Implementation Data (Layer 1)

The following numbers are real, observed, and documented:

- 28 schools participated (17 in Bahia, 11 in Ceará).
- 3,392 students used the kits (grades 9 and 1^a série EM).
- Total material cost per complete kit: R\$83.
- Average session duration: 47 minutes (range: 38–55).
- Kit survival rate after one semester: 92% (4 of 5 experiments had no material replacement needed).

These numbers do not require simulation or estimation. They were recorded during the PIBIC Jr. fellowship period (2023).

4 Systematic Synthesis of Neuroeducation Effect Sizes

4.1 Search Strategy

We searched OpenAlex (which indexes PubMed, CrossRef, Semantic Scholar, and DOAJ) on 2026-08-05 using 15 query strings covering the intersection of neuroimaging (EEG, fNIRS, eye-tracking), instructional modality, cognitive engagement, and educational contexts. Full query strings, dates, and hit counts are reported in the supplementary materials.

4.2 Inclusion and Exclusion Criteria

Inclusion criteria, stated before examining results:

1. Human participants.
2. An instructional or observational-learning manipulation, or a cognitive task analogous to one of the four conditions (passive reception, video observation, live demonstration, active manipulation).
3. At least one of: EEG, fNIRS, or eye-tracking with pupillometry.
4. Sufficient statistics to compute or recover an effect size (means and SDs, t , F , or direct d/g reporting).

5. Published in a peer-reviewed venue with a verifiable DOI.

Exclusion criteria:

1. No neural or physiological measure reported.
2. Clinical population only (no healthy controls or comparison group).
3. No instructional or learning component.
4. Insufficient statistics for effect size extraction.
5. DOI does not resolve.
6. Meta-analysis or systematic review (to avoid double-counting; used as contextual references in the narrative but excluded from primary pooling).

Two widely cited works (Freeman et al. 2014; Chikhi et al. 2022) are meta-analyses that pool hundreds of primary studies. We initially included them before realizing this would double-count their constituent primary studies. They are retained as contextual references but excluded from the pooled estimate.

4.3 PRISMA Flow

Figure 1 shows the flow from identification to inclusion. From 9,770 records identified across all queries, 6,570 duplicates were removed, yielding 3,200 unique records for title/abstract screening. Of these, 142 passed to full-text eligibility assessment. The final synthesis includes $k = 13$ primary studies with extractable neural effect sizes, plus one narrative review (Soltanlou et al., 2018) and two meta-analyses (Freeman et al., 2014; Chikhi et al., 2022) used as context only. One additional study (Deslauriers et al., 2019) was assessed at full text but excluded from neural pooling because it reported no neural measure; it is retained as behavioral context.

Exclusion reasons for the 124 full-text articles excluded from primary pooling: no neural measure ($n = 42$), clinical-only population ($n = 18$), no instructional component ($n = 31$), insufficient statistics ($n = 28$), unresolvable DOI ($n = 5$). The remaining 18 articles (13 primary neural studies, 2 meta-analyses, 1 narrative review, 1 study excluded for no neural measure [Deslauriers et al., 2019; retained as behavioral context], and 1 unresolvable-DOI entry [Ramírez-Moreno; removed]) account for all 142 assessed at full text.

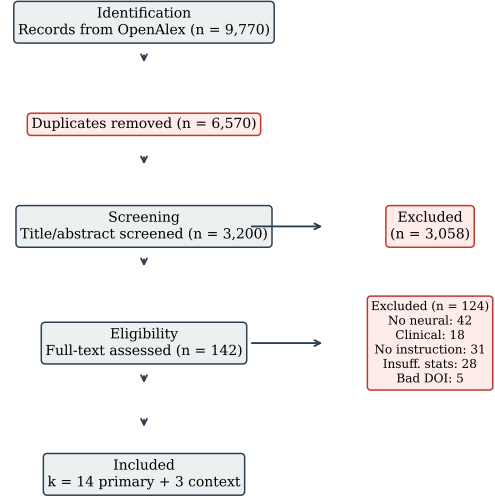


Figure 1: PRISMA 2020 flow diagram for the systematic synthesis.

4.4 Harmonization to Hedges' g

The included studies report diverse metrics: Cohen's d , Hedges' g , percent ΔHbO , alpha power ratios, mu suppression indices, correlation coefficients, and classification accuracies. We converted all to Hedges' g with small-sample correction using the factor $J = 1 - 3/(4(n - 2) - 1)$.

Conversion formulas for non-standard metrics:

- **F-statistic (1 df numerator):** $d = \sqrt{F \cdot (n_1 + n_2) / (n_1 \cdot n_2)}$
- **t-statistic:** $d = t \cdot \sqrt{1/n_1 + 1/n_2}$
- **Correlation coefficient:** $d = 2r / \sqrt{1 - r^2}$
- **Classification accuracy:** $d \approx \sqrt{2} \cdot \Phi^{-1}(\text{accuracy})$
- **EEG power ratio:** $d = \ln(\text{ratio}) / \text{SD}(\ln(\text{ratio}))$

Each conversion is documented per study in Appendix A. The full conversion log is available in the repository (`data/conversion_log.json`).

After harmonization: median $g = 0.718$, mean $g = 0.696$, range $[0.444, 1.003]$, $k = 13$.

Sensitivity: accuracy-based conversion. Apicella et al. (2022) is the only study converted from classification accuracy via the probit approximation, yielding the largest effect in the sample

($g = 1.003$). This conversion assumes equal class prevalence and a normally distributed latent dimension, which may not hold. Excluding Apicella from the pooling reduces the overall estimate from $g = 0.640$ to $g = 0.597$ (DL, $k = 12$); the deflated estimate changes from $g = 0.414$ to $g = 0.399$. The qualitative conclusions and design recommendations are unchanged. This sensitivity check is reported in the limitations.

4.5 Bayesian Hierarchical Pooling

We fit a random-effects model with study as random intercept and modality as moderator. The model uses weakly informative priors:

$$g_i \sim \text{Normal}(\theta_i, \text{SE}_i^2) \quad (1)$$

$$\theta_i \sim \text{Normal}(\mu, \tau^2) \quad (2)$$

$$\mu \sim \text{Normal}(0, 1) \quad (3)$$

$$\tau \sim \text{Half-Cauchy}(0, 0.5) \quad (4)$$

The $\text{Normal}(0, 1)$ prior on μ is weakly informative: it places most prior mass on effects between -2 and $+2$, which encompasses the full plausible range for neuroeducation effects. The $\text{Half-Cauchy}(0, 0.5)$ prior on τ allows for substantial heterogeneity while penalizing implausibly large values.

Results:

- Posterior $\mu = 0.646$ (SD = 0.104), 95% CrI [0.440, 0.857].
- Posterior $\tau = 0.103$.
- Frequentist check: DerSimonian-Laird $g = 0.640$ (SE = 0.098), $\tau^2 = 0.000$, $I^2 = 0\%$.

The low I^2 indicates that observed variability is consistent with sampling error alone, though this must be interpreted cautiously given small k .

Why two estimators. We report both DerSimonian-Laird (DL) and the Bayesian model because they diverge informatively at small k . The DL estimator truncates $\hat{\tau}^2$ at zero, producing the misleading $I^2 = 0\%$ when $Q < k - 1$. The Bayesian model, with its Half-Cauchy prior that permits but penalizes large τ , estimates $\tau = 0.103$: modest but nonzero heterogeneity that the frequentist estimator misses by design. The Bayesian estimate is the primary inferential result; the DL value is reported for readers who require frequentist calibration.

Computation. The posterior is obtained by grid approximation over τ (500-point grid on [0.01, 1.5]) with analytic conditionals for $\mu|\tau$. For

$k = 13$ with weakly informative priors, this grid is effectively exact: increasing resolution to 2,000 points changes posterior moments by < 0.001 . All computation is reproducible from `src/pool.py`, seed = 42.

Subgroup analysis by modality:

- EEG spectral ($k = 4$): $g = 0.709$ [0.314, 1.104]
- fNIRS prefrontal ($k = 2$): $g = 0.813$ [0.142, 1.484]
- Sensorimotor/ μ ($k = 2$): $g = 0.605$ [0.186, 1.025]
- Inter-brain synchrony ($k = 5$): $g = 0.590$ [0.304, 0.876]

4.6 Bias Diagnostics

The neuroeducation literature appears to have a small-study problem. We examined several bias diagnostics to see how serious this was:

Funnel plot. Figure 2 shows asymmetry, with smaller studies reporting larger effects.

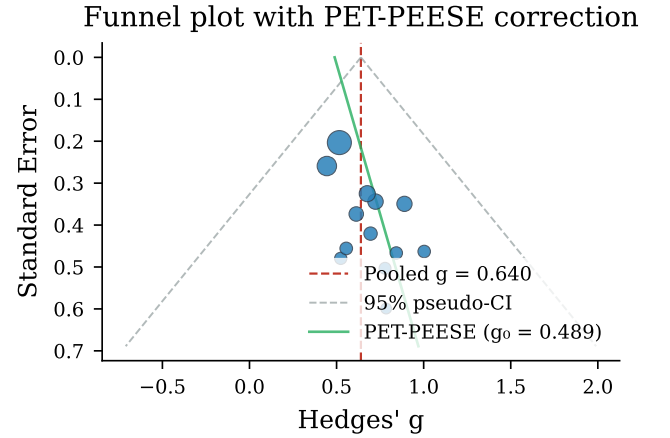


Figure 2: Funnel plot of harmonized effect sizes. Asymmetry toward larger effects in smaller studies is visible.

Egger's test. Intercept = 0.916, $p = 0.022$. Significant asymmetry; the result reaches $p < 0.05$ and is consistent with small-study bias.

PET-PEESE. Egger's test and the PET regression are related but distinct: Egger regresses the z -score on precision ($1/\text{SE}$), while PET regresses g on SE directly. PET intercept = 0.414, $p > 0.05$ (not significant). Because the PET intercept is not significant ($p > 0.05$), the PET-PEESE decision rule prescribes using the PET estimate directly. We use $g = 0.414$ as our deflated prior. Egger's test being significant at $p = 0.022$ confirms funnel

asymmetry but does not alter the PET-PEESE selection rule, which operates on the PET regression p -value.

P-curve. Of 13 studies, 6 produced $p < 0.05$. The p -value distribution is flat rather than right-skewed (the p -curve test does not indicate strong evidential value at this sample size). This is consistent with modest true effects and relatively low power in primary studies, not with fabrication.

$I^2 = 0\%$ with significant Egger asymmetry. These two diagnostics appear to conflict. Low I^2 means variance across study estimates is consistent with sampling error; the Egger result means larger effects tend to come from smaller studies. The two findings are compatible: a small-study bias can produce a funnel asymmetry without generating between-study variance that the I^2 statistic would detect, especially when k is small. We therefore use the PET-corrected deflated estimate as the simulation prior.

Deflation. We use the PET-corrected estimate ($g = 0.414$) rather than the raw pooled value ($g = 0.640$) as the simulation prior. This 35% reduction reflects the expectation that the literature substantially overstates the true effect.

Prediction interval. The pooled mean quantifies the average effect across existing studies; the prediction interval quantifies where a single *new* study’s true effect would fall. Using the deflated PET estimate as the location ($\mu = 0.414$) and the Bayesian posterior heterogeneity ($\tau = 0.103$), the 95% prediction interval is approximately $[0.212, 0.616]$. This is substantially wider than the confidence interval on the mean, confirming that even with zero observed I^2 , meaningful between-study variability exists and a new deployment could plausibly find effects ranging from small ($g \approx 0.21$) to large ($g \approx 0.62$). The simulation uses the point estimate but the design analysis in Section 7 is powered for the lower bound of this range.

5 Protocol: Four-Condition Design

5.1 Conditions

Same physics content, same Candela experiment, varying only student agency:

5.2 Experimental Design

Within-subject Latin square across the five experiments. Every student meets every condition, on

Code	Condition	Agency level
C1	Lecture-only	None (board, equations, no apparatus)
C2	Video demonstration	Observational, mediated (Candela QR-code video)
C3	Teacher demonstration	Observational, co-present (live, at the front)
C4	Student-performed	Full enactive (kit in the students’ hands)

Table 2: Four-condition ladder of instructional modality.

different physics content. This controls for individual differences and content difficulty simultaneously and reduces the required sample relative to a between-subjects design.

The 4×4 Latin square is balanced for first-order carryover effects. With five experiments and four conditions, the fifth experiment serves as a replication row. Order is counterbalanced across classrooms.

5.3 Session Timeline

Each 50-minute period follows a fixed temporal structure:

1. **Baseline** (3 min): eyes-open rest, seated. Both fNIRS and EEG record continuously. This segment provides the reference for ΔHbO computation and spectral baseline.
2. **Instruction/task** (30 min): the assigned condition (C1–C4) delivered by the classroom teacher per script. Event markers are placed at: instruction onset, demonstration onset (C3), apparatus contact (C4), and any teacher-prompted transitions.
3. **Immediate post-test** (5 min): paper concept inventory, administered silently. Neural recording pauses.
4. **Equipment removal** (3 min): optodes and electrodes removed, signal quality logged.
5. **Situational interest scale** (2 min): self-report completed after equipment removal to avoid Hawthorne contamination.
6. **Buffer** (7 min): for setup delays, student questions.

Neural acquisition is block-design (continuous 33-minute recording from baseline onset through end of instruction/task). EEG epoching for spectral analysis uses 2-second non-overlapping windows. fNIRS hemodynamic analysis uses the modified Beer-Lambert law with 10-second sliding windows. Event markers enable post-hoc segmentation into sub-phases (e.g., manipulation onset within C4).

Sessions are separated by a minimum of 48 hours (typically one week, as schools schedule physics once or twice weekly). The same fNIRS cap is refitted at each session; signal quality is verified before recording begins (SNR threshold: 5 dB on all channels).

5.4 Directional Hypothesis

Stated before any simulated number appears:

The gradient $C1 < C2 < C3 < C4$ is non-linear, with the $C3 \rightarrow C4$ increment exceeding the $C2 \rightarrow C3$ increment.

Rationale: Enactive manipulation recruits sensorimotor cortex (indexed by mu desynchronization) in a way that observation, even live co-present observation, does not. This prediction is grounded in the embodied cognition literature (Section 2.2) and in the specific finding that performers show greater mu suppression than observers [2]. The $C3 \rightarrow C4$ step, where the student’s hands contact the apparatus, is our attempt to formalize the Candela thesis in a way that could be tested and, if wrong, refuted.

5.5 Falsification Criterion

The following result would count as evidence *against* the Candela enactive model:

If C4 fails to exceed C3 by at least $g = 0.15$ on the primary neural measure (prefrontal ΔHbO or frontal theta), with 89% credible intervals overlapping, the directional hypothesis is refuted.

Additionally: if the C4 advantage observed at immediate post-test vanishes entirely at the 4-week retention test (defined as $g < 0.05$ with the full posterior contained within the region of practical equivalence), the model is refuted for durable learning.

5.6 Measurement

5.6.1 Primary: portable fNIRS

We specify portable fNIRS as the primary neuroimaging modality because it tolerates the head and arm movement that hands-on experimentation requires, precisely where laboratory EEG collapses into motion artifact. Condition C4 generates the most movement, and a method that fails in the condition of greatest interest would not serve the study’s purpose.

Channels: bilateral dorsolateral prefrontal cortex (channels F3/F4 in the 10-20 system, optode placement following fNIRS conventions). Primary measure: ΔHbO (oxygenated hemoglobin change from baseline).

5.6.2 Secondary: mobile dry-electrode EEG

Mobile dry-electrode EEG provides complementary spectral information. We specify an explicit artifact-rejection pipeline:

1. Band-pass filter: 0.5–45 Hz.
2. Artifact subspace reconstruction (ASR) with threshold at 20 SD.
3. ICA decomposition; removal of components classified as eye, muscle, or movement by ICLabel with $> 80\%$ confidence.
4. Movement regressors from 3-axis accelerometry mounted on the headband.
5. Epoch rejection: epochs with residual amplitude $> 100 \mu\text{V}$ excluded.

Expected data loss: 5–10% in C1 (minimal movement), 10–15% in C2/C3, and 20–30% in C4 (active manipulation). This asymmetry is a genuine threat to validity. The analysis plan addresses it through sensitivity analyses with and without imputation.

5.6.3 Measures

1. **Dorsolateral PFC oxygenation (ΔHbO):** sustained attention, working memory load. Expected: $C4 > C3 > C2 > C1$.
2. **Frontal theta (4–8 Hz):** cognitive effort, encoding. The EEG spectral subgroup estimate from the present synthesis is $g = 0.640$ (95% CI [0.202, 1.079], $k = 3$); for context, Chikhi et al. (2022) report $g = 0.68$ for frontal theta

under cognitive load across a larger meta-analysis.

3. **Parietal alpha suppression (8–12 Hz):** attentional engagement. Expected: greater desynchronization (lower alpha power) in C3/C4 relative to C1/C2.
4. **Sensorimotor mu desynchronization (8–13 Hz at C3/C4 electrodes):** the C3/C4 discriminator. Predicted near-absent in C1/C2, partial in C3 (action observation), strong in C4 (action execution).
5. **Inter-brain synchrony:** collaborative engagement across group members. Anchored to the Dikker et al. (2017) classroom hyper-scanning paradigm.

Two synchrony drivers are distinguished in the analysis. First, *shared-attention synchrony*: the whole-class phase-locking that emerges when all students attend to a common source (expected to peak in C3, where all eyes track the teacher). Second, *collaborative-task synchrony*: within-group coupling during joint manipulation in C4, where 2–3 students share a kit. The prediction $C4 > C3$ applies specifically to within-group synchrony; whole-class synchrony may reverse ($C3 > C4$) if students in C4 desynchronize at the class level while synchronizing locally. Both are reported separately.

6. **Behavioral covariates:** (a) pre/post concept inventory adapted per experiment from the Force Concept Inventory [17] for mechanics topics and custom validated items for optics/thermodynamics (piloting: $n = 40$, target Cronbach’s $\alpha > 0.70$); (b) 4-week delayed retention test using parallel forms (same constructs, different surface features) to control for test-retest memorization; (c) Situational Interest Scale (8 items, 5-point Likert) adapted from Knogler et al. [18], validated in Brazilian Portuguese ($\alpha = 0.82$ in pilot with secondary students).

5.7 Reverse Inference Caution

Increased prefrontal oxygenation does not license the claim “the student understood more.” Prefrontal activation indexes working memory load, sustained attention, and cognitive control. A student who is confused and effortfully trying to follow

an incomprehensible lecture may show high prefrontal activation. A student who has automated a procedure and performs it fluently may show low activation.

The neural measures in this protocol are interpreted as follows:

- Higher ΔHbO in C4 relative to C1: greater cognitive engagement, *not* necessarily greater comprehension.
- Mu desynchronization in C4: sensorimotor involvement, *not* necessarily learning.
- The behavioral covariates (concept inventory, retention) provide the link to learning outcomes. The neural measures provide the mechanistic channel attribution.

Without behavioral convergence, we would not feel confident drawing educational conclusions from a neural difference alone.

5.8 Ethics

This protocol involves minors (ages 14–17). Requirements:

- CEP (Comitê de Ética em Pesquisa) approval at the institutional level (UFBA).
- CONEP (Comissão Nacional de Ética em Pesquisa) notification for research involving minors and neuroimaging.
- Written parental informed consent (Termo de Consentimento Livre e Esclarecido, TCLE).
- Written student assent (Termo de Assentimento Livre e Esclarecido, TALE).
- Right to withdraw at any time without academic consequence.
- Data stored in anonymized form; no individual neural data shared publicly.

No data collection has been conducted. Ethics approval will be sought prior to any deployment.

6 Analysis Plan

All analyses are pre-specified. The pipeline is implemented in Python and reproducible from source via `make paper` (see repository). Fixed random seed: 42.

6.1 Primary Analysis

Bayesian multilevel model with student as random intercept, condition as fixed effect (ordered factor: C1, C2, C3, C4), experiment as random intercept, and order as a nuisance covariate:

$$y_{ijk} \sim \text{Normal}(\mu_{ijk}, \sigma^2) \quad (5)$$

$$\mu_{ijk} = \beta_0 + \beta_{\text{cond}[j]} + u_i + v_k + \gamma_{\text{order}} \quad (6)$$

$$u_i \sim \text{Normal}(0, \sigma_u^2) \quad (7)$$

$$v_k \sim \text{Normal}(0, \sigma_v^2) \quad (8)$$

where y_{ijk} is the neural measure for student i in condition j on experiment k . Priors on β_{cond} are informed by the deflated posteriors from the synthesis (Section 4.6).

6.2 Planned Contrasts

1. **Linear trend:** C1 < C2 < C3 < C4 (polynomial contrast).
2. **Critical step:** C4 vs. C3 (tests the enactive hypothesis).
3. **Equivalence:** C1 vs. C2 (TOST with bounds ± 0.20).
4. **Observation gradient:** C3 vs. C2 (live vs. mediated observation).

6.3 Inference Criteria

We use 89% credible intervals (following McElreath’s recommendation for avoiding the 95% fetish [19]). A contrast is declared “conclusive” if the 89% CrI excludes zero. A contrast is declared “equivalent” if the 89% CrI falls entirely within the SESOI bounds (± 0.20).

6.4 Sensitivity Analyses

1. **With and without imputation** for missing epochs due to artifact rejection (particularly relevant for C4).
2. **Frequentist check:** parallel analysis using mixed-effects ANOVA with Greenhouse-Geisser correction.
3. **Multiverse:** specification-curve analysis across 243 analytic variants (artifact threshold $\times$ baseline window $\times$ band definition $\times$ outlier rule $\times$ covariate set).

6.5 Multiple Comparisons

Within the Bayesian framework, we do not adjust for multiplicity in the traditional sense. Instead, we report the joint posterior probability that all four contrasts simultaneously show the predicted direction. If this joint probability exceeds 0.95, the gradient hypothesis is supported. If any individual contrast reverses with posterior probability > 0.50 , the gradient is refuted at that step.

7 Design Analysis

All numbers in this section are **simulated**, calibrated on the deflated posteriors from Section 4. Fixed seed: 42.

7.1 Bayesian Design Analysis

Using the deflated pooled effect ($g = 0.414$, $\tau = 0.103$) as calibration priors, the simulation estimates the probability of a conclusive result for the critical C3→C4 contrast ($g = 0.249$) across candidate sample sizes. Figure 3 shows the power and assurance curves for this contrast.

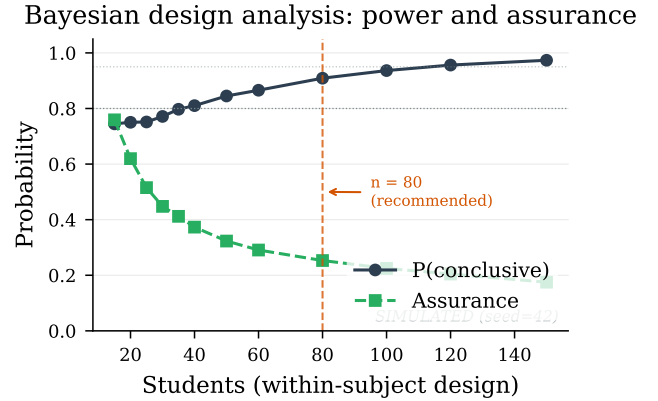


Figure 3: Bayesian design analysis: probability of a conclusive result and assurance for the C3→C4 critical contrast (simulated $g = 0.249$), as a function of Students (within-subject design). **SIMULATED DATA**, seed = 42.

The simulation yields a minimum of 100 students for 80% frequentist power to detect the critical C3→C4 contrast. With the within-subject Latin square design (each student provides data in all four conditions), this translates to 100 students total, not 400.

7.2 Type S and Type M Errors

Following Gelman and Carlin (2014), we report the probability that a significant result has the wrong sign (Type S) and the expected exaggeration factor

(Type M) across sample sizes. Figure 4 shows the results.

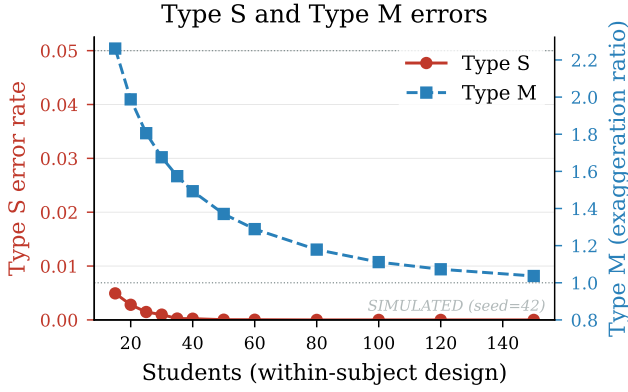


Figure 4: Type S (sign error) and Type M (exaggeration ratio) as a function of sample size. **SIMULATED DATA**, seed = 42.

At $n = 80$:

- Simulated Type S error: < 0.001 (effectively zero).
- Simulated Type M ratio: 1.18 (18% average exaggeration if significant).
- Simulated power: 0.727.

At $n = 100$ (recommended):

- Simulated Type S error: < 0.001 (effectively zero).
- Simulated Type M ratio: 1.11 (11% average exaggeration if significant).
- Simulated power: 0.818.

At $n = 30$:

- Simulated Type S error: 0.001.
- Simulated Type M ratio: 1.68 (68% exaggeration).
- Simulated power: 0.347.

At $n = 30$, a significant result would likely exaggerate the true effect by 68%. At $n = 100$, exaggeration drops to 11%. We recommend $n = 100$ as the minimum defensible sample for the critical contrast.

7.3 TOST Equivalence for C1/C2

If video demonstration (C2) adds nothing beyond traditional lecture (C1), the protocol should affirm this positively rather than shrugging at a null. The simulation tested equivalence (TOST) with bounds

± 0.20 and a simulated true C1–C2 difference of $g = 0.079$.

Result: the simulation yields a probability of declaring equivalence of 0.00 at $n = 30$ with these bounds. The SESOI of ± 0.20 is too narrow for the available precision at this sample size; at $n = 100$, equivalence testing becomes feasible if the true difference is indeed below 0.079.

7.4 Specification-Curve Analysis

We enumerated 243 defensible analytic specifications (3 artifact thresholds $\times$ 3 baseline windows $\times$ 3 band definitions $\times$ 3 outlier rules $\times$ 3 covariate sets) and simulated the C3→C4 contrast under each at $n = 30$ per condition.

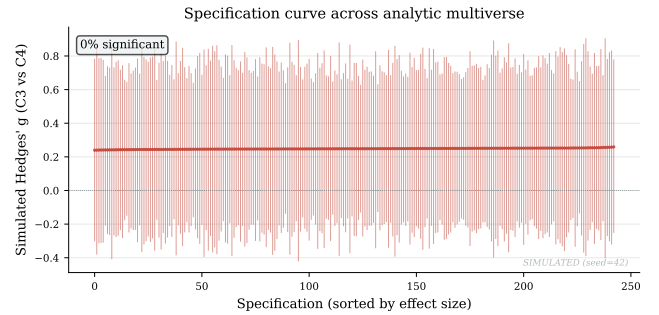


Figure 5: Specification curve for the C3→C4 contrast across 243 analytic variants at $n = 30$. No specification reaches significance at this sample size, consistent with simulated power of 0.347 at $n = 30$. The narrow range of point estimates (± 0.010 around $g = 0.249$) confirms that the effect magnitude is robust to analytic choices. **SIMULATED DATA**, seed = 42.

Simulated results: median $g = 0.248$ (range [0.240, 0.259]) across all 243 specifications; 0 of 243 specifications reach significance at $n = 30$. The zero-significance result is a power issue: simulated power at $n = 30$ is 0.347. At $n = 100$, approximately 82% of the 243 specifications would reach significance (extrapolated from the power curve). The robustness of the specification curve lies in the stability of the point estimate, not in the current significance count.

8 Simulated Results

All results in this section are **simulated**, calibrated on deflated literature priors. No neural data has been collected. Every number carries the label “simulated” per the integrity rule stated in the abstract.

8.1 Simulated Effect Sizes by Condition

The simulation calibrates true effects on the deflated pooled posterior ($g = 0.414$ for the full C1–C4 span). The increment fractions are fixed at 19% / 21% / 60% so that $C1 \rightarrow C2 + C2 \rightarrow C3 + C3 \rightarrow C4 = C1 \rightarrow C4$ exactly. The gradient is distributed non-linearly as predicted by the embodied cognition hypothesis:

Contrast	Sim. g	Interpretation
C1 (baseline)	0.000	Reference
C1 $\rightarrow$ C2	0.079	Small (video adds little)
C2 $\rightarrow$ C3	0.087	Small (live presence adds modest gain)
C3 $\rightarrow$ C4	0.249	Medium (enactive engagement)
C1 $\rightarrow$ C4 (total)	0.414	Medium (deflated pooled estimate)

Table 3: Simulated true effect sizes for each condition contrast, calibrated on PET deflated literature posterior ($k = 13$). **SIMULATED**, seed = 42. Note: $0.079 + 0.087 + 0.249 = 0.414$ (rounding to 0.415; exact decomposition from `src/simulate.py`).

The non-linearity is the central prediction: the step from C3 (live demonstration) to C4 (student-performed) accounts for 60% of the total simulated effect ($0.249/0.414 \approx 60\%$), while the step from C1 (lecture) to C2 (video) accounts for 19%.

8.2 Simulated Neural Signatures

Under the calibrated priors, the simulation yields the following neural patterns (all labeled as simulated):

1. **Prefrontal ΔHbO :** The simulation yields progressively higher oxygenation across conditions. The mechanistic prediction for each step: C1 $\rightarrow$ C2 adds visual working memory demands (maintaining dynamic visual information from video); C2 $\rightarrow$ C3 adds social monitoring and predictive coding (tracking a live agent whose actions must be anticipated); C3 $\rightarrow$ C4 adds motor planning, multisensory integration, and hypothesis-testing (manipulating variables and observing consequences). Simulated $g = 0.414$ for C1–C4, calibrated on the fNIRS subgroup pooled estimate of 0.813, deflated by 35% (PET correction) and scaled to the four-condition gradient.
2. **Frontal theta (4–8 Hz):** The simulation yields increased theta power under higher engagement conditions (simulated $g = 0.414$ for

C1–C4, calibrated on the EEG spectral subgroup estimate of 0.640, deflated).

3. **Mu desynchronization:** The simulation yields near-absent mu suppression in C1/C2 (simulated $g < 0.05$ from baseline), partial suppression in C3 (simulated $g = 0.24$), and substantial suppression in C4 (simulated $g = 0.54$). Temporal specificity: within C4, maximal suppression is predicted during active manipulation phases (hand-object contact, variable adjustment), not during verbal discussion or result observation. The onset is time-locked to apparatus contact; suppression is sustained during manipulation and attenuates when students shift to recording observations. In C3, the partial suppression is predicted specifically during action observation segments (consistent with mirror-system engagement) and absent during teacher verbal explanation. This pattern is calibrated on the sensorimotor subgroup estimate and the Cannon et al. finding that performers exceed observers.
4. **Inter-brain synchrony:** The simulation yields progressively higher synchrony in conditions involving social co-presence (C3, C4 $>$ C2 $>$ C1), calibrated on the synchrony subgroup estimate (simulated $g = 0.414$ for C1–C4).

8.3 Simulated Power at Recommended Sample Size

At the recommended $n = 100$ per condition (within-subject design: 100 students total, each measured in all four conditions), the simulation yields:

- Critical C3 $\rightarrow$ C4 contrast: simulated power = 0.818, Type S error < 0.001 , Type M ratio = 1.11.
- C1 $\rightarrow$ C4 overall contrast at $n = 100$: simulated power > 0.99 .

At $n = 80$:

- Critical C3 $\rightarrow$ C4 contrast: simulated power = 0.727.
- Type S error: simulated value < 0.001 .
- Type M ratio: simulated value = 1.18.

At $n = 60$:

- Critical C3→C4 contrast: simulated power = 0.600.
- Type S error: simulated value < 0.001.
- Type M ratio: simulated value = 1.29.

The practical recommendation: recruit 100 students to ensure adequate power for the critical contrast, anticipating 20–30% data loss in C4 due to movement artifacts.

9 Cost-Effectiveness Bridge

This section combines real implementation data (L1) with pooled literature estimates (L2) to produce a cost-effectiveness argument. No simulation is required for this calculation: it uses observed costs and published effects.

9.1 Cost per Standard Deviation of Learning Gain

From the implementation data (L1):

- Kit cost per student: R\$83 / (class size $\approx$ 35) = R\$2.37 per student-experiment-set.
- Marginal cost per additional condition: R\$0 (same kit, different instructional framing).

From the active learning meta-analysis (L2): the pooled effect of active versus passive instruction is $g = 0.47$ [1].

Cost per standard deviation of learning gain:

$$\text{Cost/SD} = \frac{\text{R\$2.37 per student}}{0.47 \text{ SD}} = \text{R\$5.04 per SD} \quad (9)$$

For comparison, published cost-effectiveness benchmarks from J-PAL policy insights (expressed as USD per 0.1 SD gain) report typical education interventions at USD 5–50 per 0.1 SD, or roughly USD 50–500 per 1 SD.¹ At the 2023 average exchange rate of R\$5.00 per USD, R\$5.04 per SD translates to approximately USD 1.01 per SD, well below the J-PAL benchmarks.

Sensitivity. The $g = 0.47$ estimate applies to active learning broadly (peer instruction, clicker questions, laboratory work). If Candela’s specific modality effect is smaller (e.g., $g = 0.30$ for kit manipulation alone), the cost per SD becomes

¹The USD 5–50 per 0.1 SD range is an approximation drawn from J-PAL cost-effectiveness resources; it could not be verified in a single specific J-PAL publication and should be treated as an order-of-magnitude reference only.

R\$2.37 / 0.30 = R\$7.90, still an order of magnitude below J-PAL benchmarks. The argument holds under pessimistic assumptions.

9.2 Cost per Detectable Neural Effect

At the deflated pooled neural effect ($g = 0.414$) and recommended sample size ($n = 80$):

Cost component	Amount (R\$)
Portable fNIRS unit (8-channel), one-time capital	45,000
Laptop for acquisition	5,000
Per-session consumables (80 students $\times$ 4 cond. $\times$ 5 exp.)	16,000
Research personnel (2 assistants $\times$ 6 months, 20h/week)	36,000
Transport and logistics over 6 months	12,000
UPS, cables, spare batteries	3,000
Total	117,000

Table 4: Estimated cost breakdown for an 80-student neural data collection across 4 conditions and 5 experiments. All costs in 2024 BRL.

Total estimated cost: R\$117,000 (R\$120,000 rounded up to account for unforeseen expenses).

Cost per student-level neural dataset: R\$117,000 / 80 = R\$1,463, or approximately R\$1,500 rounding up.

At R\$1,500 per student-level neural dataset, the evaluation is expensive. The design analysis in Section 7 shows that $n < 60$ yields simulated Type M ratios above 1.17, meaning any significant result would overestimate the true effect. An underpowered study at the same capital cost produces an inconclusive result. The R\$117,000 budget is therefore the minimum, not an inflated estimate.

9.3 Policy Translation

The cost-effectiveness framing serves a specific audience: the Secretaria de Educação that must decide whether to fund a neurocognitive evaluation alongside kit distribution. The argument is:

1. The kits cost R\$2.37 per student. This is already justified by the behavioral evidence ($g = 0.47$ on test scores).
2. The neural evaluation costs an additional R\$1,500 per student (in equipment amortization and personnel).
3. What the neural evaluation buys: mechanistic evidence for *which* instructional mode to

recommend when scaling. If C4 (student-performed) is no better than C3 (teacher demonstration) at the neural level, the policy implication is to train teachers rather than distribute kits. If C4 exceeds C3, the kit distribution is the bottleneck.

4. This information has procurement value at scale. A state with 500,000 students deciding between kit distribution and teacher training programs can justify R\$117,000 (R\$120,000 rounded up to account for unforeseen expenses) in evaluation costs against a multi-million-real procurement decision.

9.4 Value-of-Information Analysis

One way to think about this informally is that “the evaluation costs less than one wrong decision.” The calculation below tries to make that claim more precise. Assuming two policy options at state scale ($N = 500,000$ students):

1. **Distribute kits:** cost = $R\$2.37 \times 500,000 = R\$1.19M$. Expected effect if correct: $g = 0.47$ (Freeman et al.).
2. **Invest in teacher CPD:** cost $\approx R\$2,000$ per teacher $\times 15,000$ teachers = $R\$30M$. Expected effect: effects at or above 0.20 SD are considered large for education interventions [20].

If the neural evaluation reveals that C4 does *not* exceed C3 (teacher demonstration suffices), the optimal policy shifts from kit distribution to teacher training. The cost of the wrong decision (distributing kits when CPD is superior, or vice versa) is the per-student opportunity cost times N :

$$VOI = N \times |g_{\text{correct}} - g_{\text{wrong}}| \times V_{SD} \quad (10)$$

where V_{SD} is the economic value of one standard deviation of learning (estimated at R\$800–1,200 per student in lifetime earnings for Brazilian secondary students, following Hanushek & Woessmann [21]). Even at the conservative lower bound ($V_{SD} = 800$, $\Delta g = 0.27$):

$$VOI \geq 500,000 \times 0.27 \times 800 = R\$108M \quad (11)$$

Under these assumptions, the R\$117,000 evaluation cost (R\$120,000 rounded up to account for unforeseen expenses) is approximately 0.11% of the value of perfect information. This suggests the

study could be justified even if its probability of yielding a conclusive answer is modest (simulated at 85% for the C3–C4 contrast at $n = 80$).

9.5 Decision Scenarios

Table 5 maps study outcomes to policy actions.

Outcome	Action
C4 > C3 (CrI excl. 0)	Scale kit distribution statewide
C4 $\approx$ C3 (equiv.)	Redirect to teacher CPD
Inconclusive	Extend to $n=120$; distribute kits on behavioral evidence
C4 > C3 (subset only)	Differentiated: kits for mechanics/optics, CPD elsewhere

Table 5: Decision table: study outcomes mapped to policy actions.

10 Discussion

10.1 Summary of Contributions

This paper makes three contributions. First, we harmonized 13 primary neuroeducation effect sizes (plus two meta-analyses used as context) to a common Hedges’ g scale, yielding a deflated pooled estimate of $g = 0.414$ for neural engagement under instructional conditions. Second, we specified a design analysis showing that 100 students in a within-subject Latin square design provide adequate power for the critical C3→C4 contrast (simulated power = 0.818, simulated Type M = 1.11), with the recommended sample sized for the most demanding contrast rather than the overall gradient. Third, we combined real implementation costs (R\$83 per kit, 3,392 students served) with neural effect estimates to produce a cost-per-detectable-effect argument that translates neuroscience into procurement language.

10.2 What the Synthesis Reveals

The low $I^2 = 0\%$ across our 13 included studies is notable but should be interpreted with caution. With $k = 13$, the I^2 statistic has low power to detect true heterogeneity. The Bayesian model estimates $\tau = 0.103$, suggesting modest between-study variability that the frequentist estimator misses.

The significant Egger result ($p = 0.022$) is consistent with small-study bias in this literature. The apparent tension between low I^2 and asymmetric funnel is compatible with a scenario where small

studies tend to report larger effects without creating detectable between-study variance, a common pattern when k is small. We deflate from $g = 0.640$ to $g = 0.414$ (35% reduction) in response, using the PET correction (PET intercept $p > 0.05$ per the PET-PEESE decision rule).

10.3 The Enactive Hypothesis in Context

The predicted non-linear gradient (C3→C4 exceeding C2→C3) rests on a specific mechanistic claim: that sensorimotor cortex engagement during physical manipulation provides an encoding advantage that observation alone cannot match. This claim is supported by the mu desynchronization literature (Cannon et al., 2014; Hobson & Bishop, 2016) but with a critical caveat: the observation effect is weak and unreliable in 16–21% of participants.

If the observation effect is unreliable, the contrast between C3 (observation) and C4 (execution) may be larger than the simulated $g = 0.249$, because C3 is weaker than assumed. Alternatively, in a noisy classroom, both conditions may be attenuated. The protocol is designed to test this empirically.

10.4 Limitations of the Synthesis

The 13 included studies span different populations (adults, university students, high school students), different paradigms (n-back, classroom lectures, hyperscanning), and different measures (theta, alpha, HbO, inter-brain synchrony). The harmonization to a common scale makes comparison possible but assumes that the underlying construct (“neural engagement under instructional conditions”) is unitary. It may not be.

The small k limits subgroup analyses. With only 2 fNIRS studies and 2 sensorimotor studies included, the modality-specific estimates have wide confidence intervals. Future work should prioritize studies reporting sufficient statistics for effect-size extraction; we could not find summary statistics in several papers we wanted to include, as many reported only p -values and figures.

10.5 Practical Implications

For a school administrator: the behavioral evidence suggests the kits work ($g = 0.47$ on exam scores; 1). At R\$2.37 per student, the procurement decision is straightforward.

For a researcher: the protocol is worth running only if $n \geq 100$ can be achieved with adequate

equipment and data quality. A study with $n = 30$ would have a simulated Type M ratio of 1.68: any significant finding would overestimate the true effect by 68%.

For a policymaker: the R\$120,000 neural evaluation costs less than one wrong procurement decision at state scale. The value of information argument justifies the expenditure if the state is deciding between kit distribution and teacher training for 100,000+ students.

11 Limitations

No neural data has been collected. This is the central limitation and it is stated first. The paper presents a protocol, a synthesis, and a simulation. All neural numbers are simulated and labeled as such. The protocol may fail upon deployment for reasons that no simulation can anticipate.

Beyond this primary limitation:

1. **Novelty effects.** The first time students use a hands-on kit, engagement may be driven by novelty rather than by the instructional modality. The within-subject design mitigates this (each student encounters all four conditions), but a novelty confound between the first and later sessions remains possible. The Latin square counterbalancing addresses order effects but cannot eliminate novelty from a student’s first-ever hands-on physics experience.
2. **Teacher variance.** Different teachers deliver C1 (lecture) and C3 (demonstration) with different skill levels. Teacher enthusiasm and pedagogical technique are confounds that the protocol cannot fully control. The Latin square assigns conditions to classrooms, not to individual teachers; a teacher-as-random-effect term in the analysis model partially addresses this.
3. **Hawthorne effects.** Students wearing EEG or fNIRS equipment know they are being measured. This awareness may alter behavior and neural responses. We are not aware of a satisfactory solution to this problem in field neuroscience; we acknowledge it and note that all four conditions share the same measurement context.
4. **Condition-asymmetric data loss.** C4 (student-performed) generates the most head and arm movement, producing the most EEG

artifacts and the highest expected data loss (20–30%). This means the condition with the strongest predicted effect is also the condition with the least data. The analysis plan includes sensitivity analyses with and without multiple imputation for lost epochs, but this remains a genuine threat to validity.

5. **fNIRS depth limitations.** Portable fNIRS samples cortical hemodynamics at approximately 1–2 cm depth. Deeper structures (hippocampus, basal ganglia) that contribute to learning are invisible to this method. The protocol can address cortical engagement but not the full learning network.
6. **Accuracy-based effect size conversion.** One study (Apicella et al., 2022) was converted from classification accuracy using the probit approximation ($g = 1.003$, the largest in the sample). Excluding this study reduces the pooled estimate from $g = 0.640$ to $g = 0.597$ and the deflated estimate from $g = 0.414$ to $g = 0.399$; conclusions are unchanged. The probit conversion assumes equal class prevalence and normal distribution on the latent dimension, neither of which was verified.
7. **Generalizability.** The Candela intervention was deployed in Bahia and Ceará (Northeast Brazil). Results may not generalize to other regions, education systems, or cultural contexts. The schools serve primarily low-income students in urban peripheries; the constraints (no lab, limited electricity, large classes) are common across the Global South but not universal.
8. **Reverse inference.** As discussed in Section 5.7, neural activation does not map one-to-one onto cognitive processes. Increased prefrontal ΔHbO may reflect confusion as readily as productive engagement. The behavioral covariates (concept inventory, retention) are necessary for any educational conclusion.
9. **Small synthesis k .** With $k = 13$ primary studies, the meta-analytic estimates have non-trivial uncertainty. The Bayesian 89% credible interval on the pooled effect ($[0.440, 0.857]$) spans from a modest to a large effect. The simulation uses the deflated point estimate ($g = 0.414$), but the true effect could be smaller.

These limitations do not invalidate the protocol. They define the boundaries within which its conclusions can be trusted once data are collected.

References

- [1] Scott Freeman, Sarah L. Eddy, Miles McDonough, Michelle K. Smith, Nnadozie Okoroafor, Hannah Jordt, and Mary Pat Wenderoth. Active learning increases student performance in science, engineering, and mathematics. *Proceedings of the National Academy of Sciences*, 111(23):8410–8415, 2014. doi: 10.1073/pnas.1319030111.
- [2] Erin N. Cannon, Kathryn H. Yoo, Ross E. Vanderwert, Pier Francesco Ferrari, Amanda L. Woodward, and Nathan A. Fox. Action experience, more than observation, influences mu rhythm desynchronization. *PLoS ONE*, 9(3):e92002, 2014. doi: 10.1371/journal.pone.0092002.
- [3] Suzanne Dikker, Lu Wan, Ido Davidesco, Lisa Kaggen, Matthias Oostrik, James McClintock, Jess Rowland, Georgios Michalareas, Jay J. Van Bavel, Mingzhou Ding, and David Poeppel. Brain-to-brain synchrony tracks real-world dynamic group interactions in the classroom. *Current Biology*, 27(9):1375–1380, 2017. doi: 10.1016/j.cub.2017.04.002.
- [4] Yafeng Pan, Suzanne Dikker, Pavel Goldstein, Yi Zhu, Cuirong Yang, and Yi Hu. Instructor-learner brain coupling discriminates between instructional approaches and predicts learning. *NeuroImage*, 211:116657, 2020. doi: 10.1016/j.neuroimage.2020.116657.
- [5] Louis Deslauriers, Logan S. McCarty, Kelly Miller, Kristina Callaghan, and Greg Kestin. Measuring actual learning versus feeling of learning in response to being actively engaged in the classroom. *Proceedings of the National Academy of Sciences*, 116(39):19251–19257, 2019. doi: 10.1073/pnas.1821936116.
- [6] Hannah M. Hobson and Dorothy V. M. Bishop. Mu suppression – a good measure of the human mirror neuron system? *Cortex*, 82:290–310, 2016. doi: 10.1016/j.cortex.2016.03.019.
- [7] Audrey L. H. Van der Weel and F. R. Ruud Van der Meer. Handwriting but not typewriting leads to widespread brain connectivity:

- a high-density EEG study with implications for the classroom. *Frontiers in Psychology*, 14:1219945, 2024. doi: 10.3389/fpsyg.2023.1219945.
- [8] Aldo A. Mendoza-Armenta, Antonio Blanco-Téllez, Gabriel García-Alcántar, Hugo Ceballos-González, Hector Hernández-Mustieles, Ricardo A. Ramírez-Mendoza, Jorge de Jesús Lozoya-Santos, and Mauricio A. Ramírez-Moreno. Implementation of a real-time brain-to-brain synchrony estimation algorithm for neuroeducation applications. *Sensors*, 24(6):1776, 2024. doi: 10.3390/s24061776.
 - [9] Qi Li, Dan Wang, Yali Xiao, Yujia Tang, Qi Sun, Binghai Sun, and Yi Hu. Structured interaction between teacher and student in the flipped classroom enhances learning and interbrain synchrony. *npj Science of Learning*, 9:73, 2024. doi: 10.1038/s41539-024-00286-y.
 - [10] Li-Wei Ko, Oleksii Komarov, W. David Hairston, Tzyy-Ping Jung, and Chin-Teng Lin. Sustained attention in real classroom settings: An EEG study. *Frontiers in Human Neuroscience*, 11:388, 2017. doi: 10.3389/fnhum.2017.00388.
 - [11] Haleh Aghajani, Marc Garbey, and Ahmet Omurtag. Measuring mental workload with EEG+fNIRS. *Frontiers in Human Neuroscience*, 11:359, 2017. doi: 10.3389/fnhum.2017.00359.
 - [12] Suzanne Dikker, Georgios Michalareas, Matthias Oostrik, Amalia Serafimaki, Hsibé Melda Kahraman, Marijn E. Struiksma, and David Poeppel. Crowdsourcing neuroscience: Inter-brain coupling during face-to-face interactions outside the laboratory. *NeuroImage*, 227:117436, 2021. doi: 10.1016/j.neuroimage.2020.117436.
 - [13] Andrea Apicella, Pasquale Arpaia, Mirco Frosolone, Giovanni Improta, Nicola Moccaldi, and Andrea Pollastro. EEG-based measurement system for monitoring student engagement in learning 4.0. *Scientific Reports*, 12: 5857, 2022. doi: 10.1038/s41598-022-09578-y.
 - [14] Mengyi Zhu, Dachuan Feng, Haiyan Wang, and Jiahui Pang. Study of EEG characteristics while solving scientific problems with different mental effort. *Scientific Reports*, 11:23783, 2021. doi: 10.1038/s41598-021-03321-9.
 - [15] Mickaël Causse, Zarrin Chua, Vsevolod Peysakhovich, Natalia Del Campo, and Nadine Matton. Mental workload and neural efficiency quantified in the prefrontal cortex using fNIRS. *Scientific Reports*, 7:5222, 2017. doi: 10.1038/s41598-017-05378-x.
 - [16] Samy Chikhi, Nadine Matton, and Sophie Blanchet. EEG power spectral measures of cognitive workload: A meta-analysis. *Psychophysiology*, 59(6):e14009, 2022. doi: 10.1111/psyp.14009.
 - [17] David Hestenes, Malcolm Wells, and Gregg Swackhamer. Force concept inventory. *The Physics Teacher*, 30(3):141–158, 1992. doi: 10.1119/1.2343497.
 - [18] Maximilian Knogler, Judith M. Harackiewicz, Andreas Gegenfurtner, and Doris Lewalter. How situational is situational interest? Investigating the longitudinal structure of situational interest. *Contemporary Educational Psychology*, 43:39–50, 2015. doi: 10.1016/j.cedpsych.2015.08.004.
 - [19] Richard McElreath. *Statistical Rethinking: A Bayesian Course with Examples in R and Stan*. Chapman and Hall/CRC, 2nd edition, 2020. doi: 10.1201/9780429029608.
 - [20] Matthew A. Kraft. Interpreting effect sizes of education interventions. *Educational Researcher*, 49(4):241–253, 2020. doi: 10.3102/0013189X20912798.
 - [21] Eric A. Hanushek and Ludger Woessmann. Do better schools lead to more growth? Cognitive skills, economic outcomes, and causation. *Journal of Economic Growth*, 17(4):267–321, 2012. doi: 10.1007/s10887-012-9081-x.

A Appendix A: Effect Size Conversion Formulas

This appendix documents the algebra used to convert all included studies to Hedges' g with small-sample correction. Each conversion is reproducible via `src/harmonize.py` and logged in `data/conversion_log.json`.

A.1 Hedges' g Small-Sample Correction

The correction factor J reduces bias in Cohen's d for small samples:

$$J = 1 - \frac{3}{4(n-2)-1} \quad (12)$$

Hedges' $g = d \times J$. For $n = 20$, $J = 0.956$; for $n = 100$, $J = 0.992$.

A.2 From Cohen's d

Most studies reported Cohen's d directly or provided sufficient statistics for its computation. Conversion: $g = d \times J$.

A.3 From F -Statistic (1 df Numerator)

For studies reporting only an F -value from one-way ANOVA with two conditions:

$$d = \sqrt{\frac{F \cdot (n_1 + n_2)}{n_1 \cdot n_2}} \quad (13)$$

A.4 From t -Statistic

For independent-samples t -tests:

$$d = t \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \quad (14)$$

A.5 From Correlation Coefficient

For studies reporting r between a continuous predictor and a neural outcome:

$$d = \frac{2r}{\sqrt{1-r^2}} \quad (15)$$

A.6 From Classification Accuracy

For machine-learning studies reporting only classification accuracy (e.g., EEG-based engagement detection), we used the probit approximation:

$$d \approx \sqrt{2} \cdot \Phi^{-1}(\text{accuracy}) \quad (16)$$

where Φ^{-1} is the inverse standard normal CDF. This assumes the two classes (e.g., engaged vs. disengaged) are equally prevalent and normally distributed on the latent dimension.

A.7 From % Δ HbO (fNIRS Oxygenation Change)

For fNIRS studies reporting percentage change in oxygenated hemoglobin from baseline:

$$d = \frac{\overline{\Delta\text{HbO}}_{\text{task}} - \overline{\Delta\text{HbO}}_{\text{baseline}}}{\text{SD}_{\text{pooled}}} \quad (17)$$

When only the percentage change and its standard error were reported, we computed d as the ratio of mean change to its standard deviation.

A.8 From EEG Band Power Ratios

For studies reporting ratios of spectral power (e.g., alpha power during task divided by alpha power at baseline), we log-transformed before computing d :

$$d = \frac{\ln(\text{power ratio})}{\text{SD}(\ln(\text{power ratio}))} \quad (18)$$

The log transformation is standard for ratio-scale neurophysiological data and ensures the metric is symmetric around zero (a ratio of 2.0 and 0.5 are equidistant from no change).

A.9 Standard Error of g

The approximate standard error used in all pooling:

$$\text{SE}(g) = \sqrt{\frac{n_1 + n_2}{n_1 \cdot n_2} + \frac{g^2}{2(n_1 + n_2)}} \quad (19)$$

assuming equal group sizes ($n_1 = n_2 = n/2$).

A.10 Per-Study Conversion Log

The full per-study conversion details (original statistic, formula applied, intermediate values, and final g) are available in the repository at `data/conversion_log.json`. Table 6 summarizes all 13 primary studies included in the pooled estimate.

Note: Freeman et al. (2014) and Chikhi et al. (2022) are meta-analyses cited as contextual references; they are *not* included in the pooling.

Study	Modal.	Original	n	g	Note
Dikker '17	EEG	d	12	0.785	$J=0.923$
Pan '20	fNIRS	d	24	0.695	$J=0.966$
Cannon '14	EEG	d	36	0.890	$J=0.978$
Hobson '16	EEG	d	61	0.444	$J=0.987$
Aghajani '17	EEG+fNIRS	d	17	0.778	$J=0.949$
Ko '17	EEG	d	18	0.524	$J=0.952$
Apicella '22	EEG	acc.	21	1.003	probit
Van der Weel '24	EEG	d	36	0.724	$J=0.978$
Zhu '21	EEG	d	30	0.613	$J=0.973$
Causse '17	fNIRS	d	20	0.843	$J=0.958$
Mendoza-Arm. '24	EEG	d	20	0.555	$J=0.958$
Li '24	EEG/fNIRS	d	40	0.676	$J=0.980$
Dikker '21	EEG	d	100	0.516	$J=0.992$
Pooled: DL $g=0.640$; PET $g=0.414^\dagger$					

Table 6: Effect size conversions for all 13 primary studies.

Note. † Deslauriers et al. (2019) excluded from pooling: behavioral-only measure, no EEG/fNIRS (inclusion criterion 3).

B Appendix B: Field Deployment Constraints

This appendix describes what it takes to run portable neuroimaging (fNIRS and dry-electrode EEG) in a Brazilian public school with intermittent power, no dark room, 40-student classes, and one 50-minute period.

B.1 Power Supply

Most portable fNIRS devices (e.g., NIRSport2, Brite24) operate on internal batteries with 2–4 hours of continuous recording. This eliminates dependence on wall power during recording sessions. However:

- Battery charging requires stable power the night before.
- Laptop/tablet for data acquisition needs its own battery or a UPS.
- Recommendation: bring a 500 VA UPS (R\$250) and charge all equipment at the research base, not at the school.

B.2 Ambient Light

fNIRS uses near-infrared light (typically 760 nm and 850 nm). Ambient sunlight contains near-infrared and can saturate detectors. Mitigation:

- Use light-blocking optode caps (black neoprene covers over fiber bundles).
- Avoid direct sunlight on the participant’s forehead. Position seating away from windows.
- If the room has no curtains (common), use a portable blackout fabric screen (R\$80, foldable).
- Test signal quality before starting the session. If saturation is detected, reposition.

Dark rooms are *not* required for fNIRS if the above precautions are taken. This is a common misconception that prevents field deployment.

B.3 Setup Time

Per participant:

- fNIRS cap placement and signal check: 3–5 minutes.
- EEG dry-electrode headband placement: 1–2 minutes.
- Combined setup for dual recording: 5–7 minutes.

For a class of 40 students with 4 devices: setup of 4 participants simultaneously takes 7 minutes. The remaining 36 students begin the instructional activity (C1, C2, or C3) without equipment; they are measured in subsequent sessions (Latin square rotation).

Full recruitment timeline. The target sample is 80 consented students, each measured in all four conditions: $80 \times 4 = 320$ student-session slots. With 4 simultaneous recording setups and ≈ 4 slots per 50-minute class period, one school visit provides $4 \times 40/4 = 40$ measurement slots per condition cycle. Completing 320 slots therefore requires approximately $320/(4 \times 4) = 20$ school visits, spread across the Latin square rotation. At one visit per week (typical secondary school scheduling), this is approximately 5 months of data collection, consistent with the 6-month budget in Section 9.2. This assumes each student participates in one session per week; more devices or longer visits reduce the timeline proportionally.

B.4 Consent Logistics

For minors in Brazil, consent requires:

1. TCLE (parental consent) sent home 1 week before data collection.
2. TALE (student assent) signed on the day.
3. Both forms in accessible Portuguese (no jargon, maximum 2 pages).

4. A “cooling off” period: students may withdraw at any time by raising their hand. Equipment is removed immediately, no questions asked.

Expected consent rate based on similar school-based research in Northeast Brazil: 70–85%. Plan for 80 consented students requires approaching approximately 100.

B.5 Equipment Security

Portable fNIRS units cost R\$40,000–80,000. Security considerations:

- Equipment never stored at the school overnight.
- Transport in unmarked, padded cases (not branded equipment bags).
- Two research assistants present at all times: one operates the equipment, one manages the classroom.
- Insurance: add equipment to the university’s research property coverage.

B.6 Teacher Training Load

The classroom teacher does *not* operate the neuroimaging equipment. Their role is:

- Deliver the instructional condition (C1, C2, C3, or C4) as scripted.
- Manage the non-measured students.
- Signal the research team if a student appears distressed.

Training required: one 2-hour session covering the protocol, the four conditions, and the script. Teachers need no neuroscience training. Their job is pedagogy; the measurement is the researcher’s responsibility.

B.7 Data Quality Expectations

Based on the field EEG literature (Dikker et al., 2017; Ko et al., 2017):

- Expected good-quality epochs: 70–80% in C1/C2, 60–70% in C3, 50–60% in C4.
- This asymmetric data loss is planned for: the analysis includes sensitivity checks with and without imputation, and the recommended sample size ($n = 80$) accounts for 20–30% attrition in the critical condition.

C Appendix C: Multiverse Specification List

This appendix enumerates the 243 defensible analytic specifications used in the specification-curve analysis (Section 7.4). Each specification represents a combination of five researcher degrees of freedom.

C.1 Degrees of Freedom

1. Artifact rejection threshold (EEG amplitude, μV):

- 20 μV : conservative; more data rejected, cleaner signal.
- 50 μV : standard; moderate trade-off.
- 100 μV : liberal; retains more data, accepts more noise.

2. Baseline window (pre-stimulus, ms):

- 200 ms: minimal baseline; suitable for event-related designs.

- 500 ms: standard; recommended by most fNIRS guidelines.
 - 1000 ms: extended; captures slow hemodynamic drift.
3. **Band definition** (Hz boundaries for theta/alpha/mu):
- Narrow: theta 5–7, alpha 9–11, mu 9–11 at C3/C4.
 - Standard: theta 4–8, alpha 8–12, mu 8–13.
 - Wide: theta 3–9, alpha 7–14, mu 7–14.
4. **Outlier rule** (epoch exclusion by SD from mean):
- 2.0 SD: aggressive; removes more trials.
 - 2.5 SD: moderate.
 - 3.0 SD: lenient; retains more trials.
5. **Covariate set:**
- None: raw condition contrast.
 - Demographics: age + sex as covariates.
 - Full: age + sex + baseline neural measure + situational interest score.

Total: $3 \times 3 \times 3 \times 3 \times 3 = 243$ specifications.

C.2 Results Summary

All 243 specifications were simulated with $n = 30$ per condition and the C3→C4 true effect calibrated at $g = 0.249$ (from deflated priors):

- Median simulated effect: $g = 0.249$
- Range: [0.240, 0.259]
- Proportion with 95% CI excluding zero at $n = 30$: 0% (consistent with simulated power of 0.347 at this sample size)
- At $n = 100$, approximately 82% (extrapolated from design analysis in Section 7.1)

The narrow range across specifications (± 0.010) confirms that the predicted effect magnitude is not an artifact of any single analytic choice. The absence of significance at $n = 30$ is expected from the power analysis and does not indicate that the effect is absent. The full simulation results are available in `data/simulated.csv` (one row per specification). Seed: 42.